

Deciphering the Diagenetic History of the Avon Park Formation through a Trace Metal Perspective

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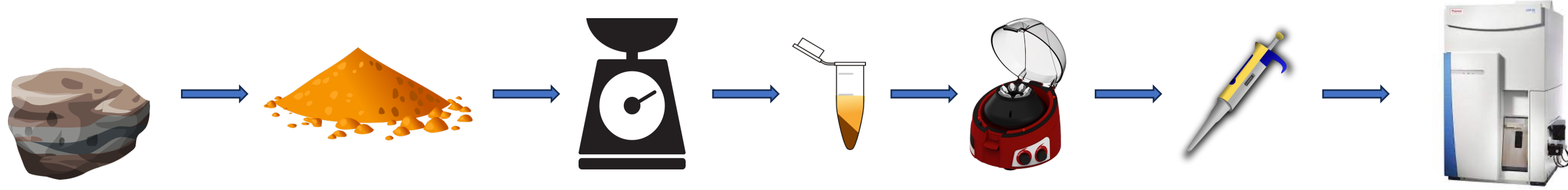


Background

- Shallow marine carbonates are invaluable archives of past seawater chemistry and environmental conditions.
- They are highly susceptible to diagenetic changes, which can significantly impact the accuracy of past seawater reconstructions, in this study we focus primarily on dolomites.
- To comprehensively assess these diagenetic processes, this study focuses on the mid-Eocene Avon Park Formation located in Florida.
- Trace metal abundances are explored to gain insights into biogeochemical reactions, redox shifts, and the influence of biotic components within the sediment.
- Early marine diagenesis has led to partial dolomitization of these shallow-water carbonates, significantly affecting their observed geochemistry.

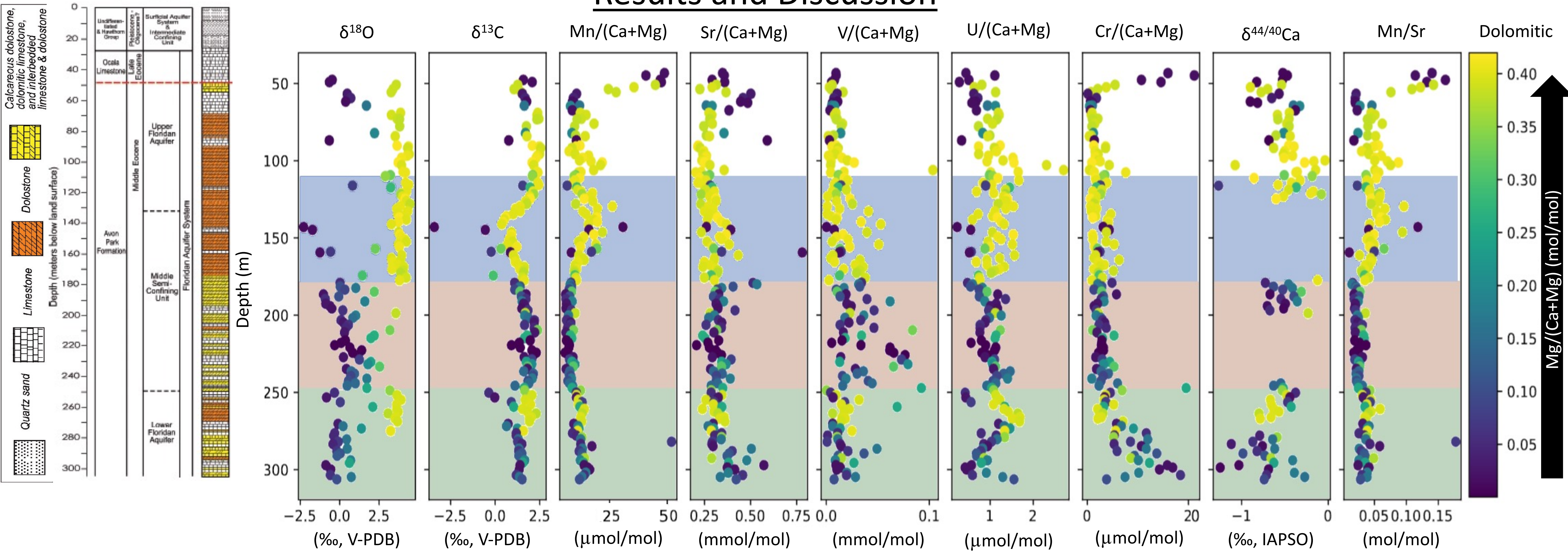
Methods

- Samples were selected to capture the Early-Middle Eocene.
- Samples were powdered and portioned into 10 mg each, then dissolved, oxidatively cleaned, and matrix matched to then be ran in the iCAP RQ (ICP-MS)
- Combined trace element concentrations with a suite of previously conducted isotopic analyses ($\delta^{13}\text{C}$, $\delta^{18}\text{O}$, $\delta^{44/40}\text{Ca}$).
- Conducted detailed trace element analysis, measuring Mg, Sr, U, V, Cr, Mn, and Fe.

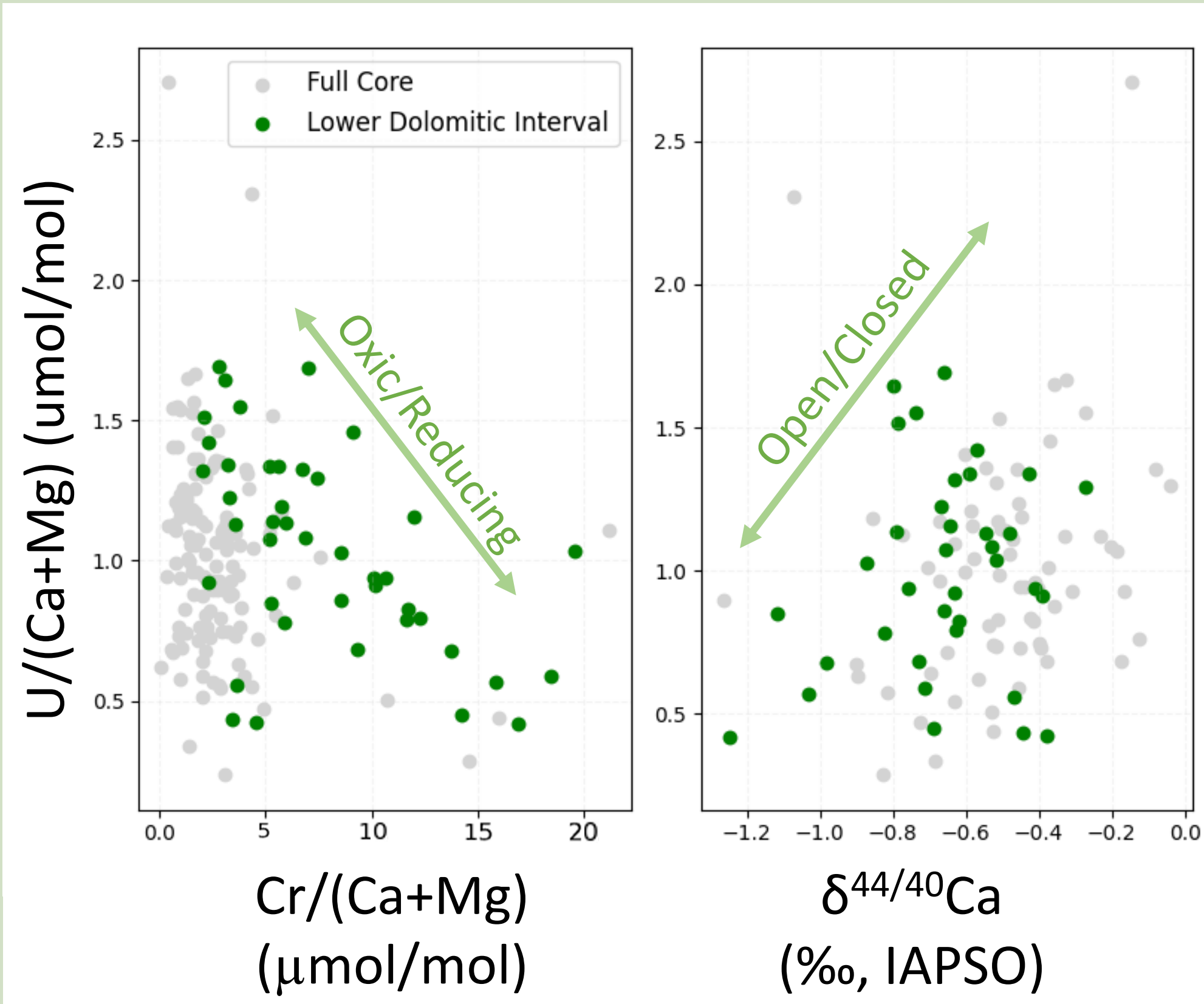


[5]

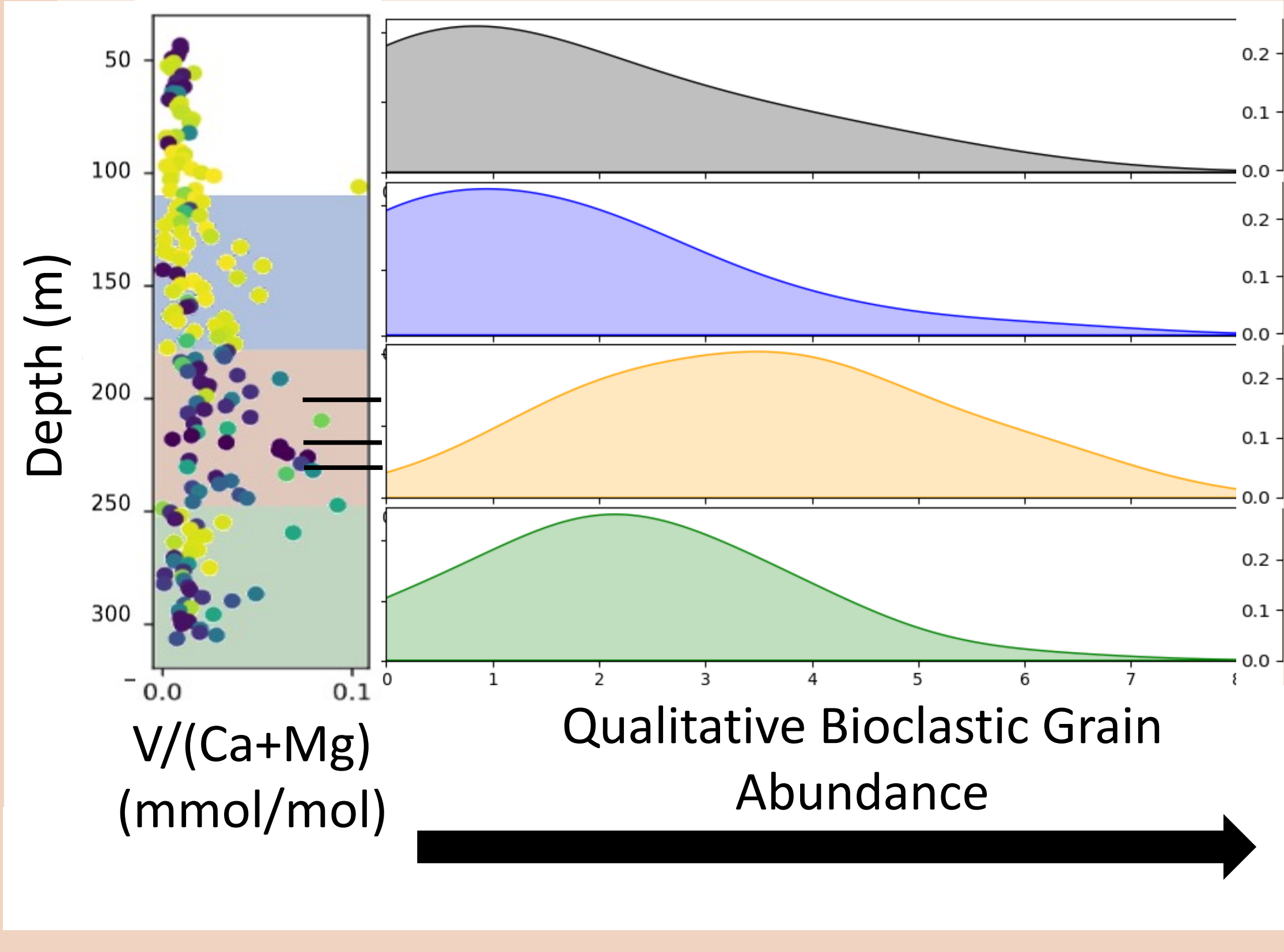
Results and Discussion



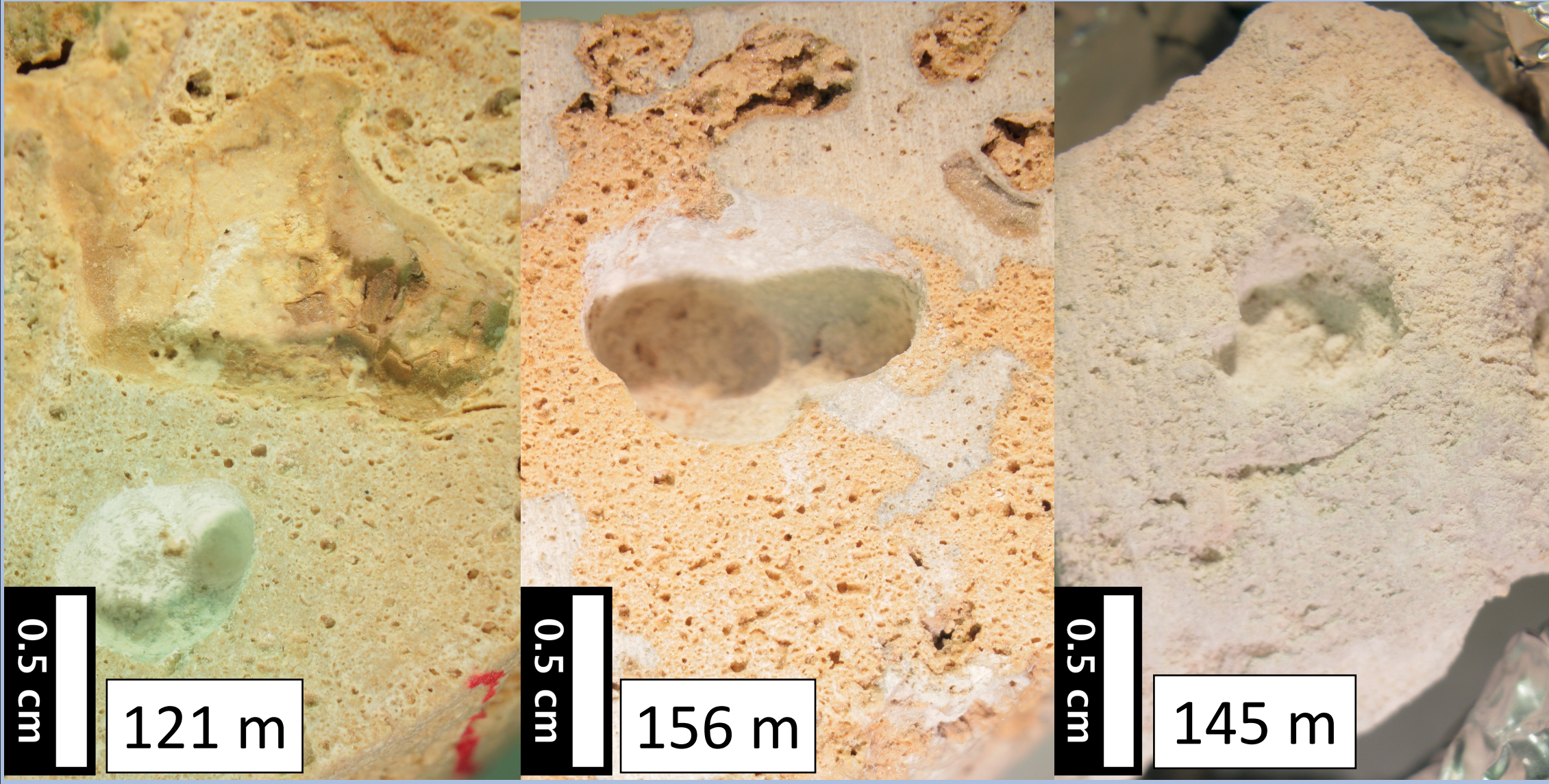
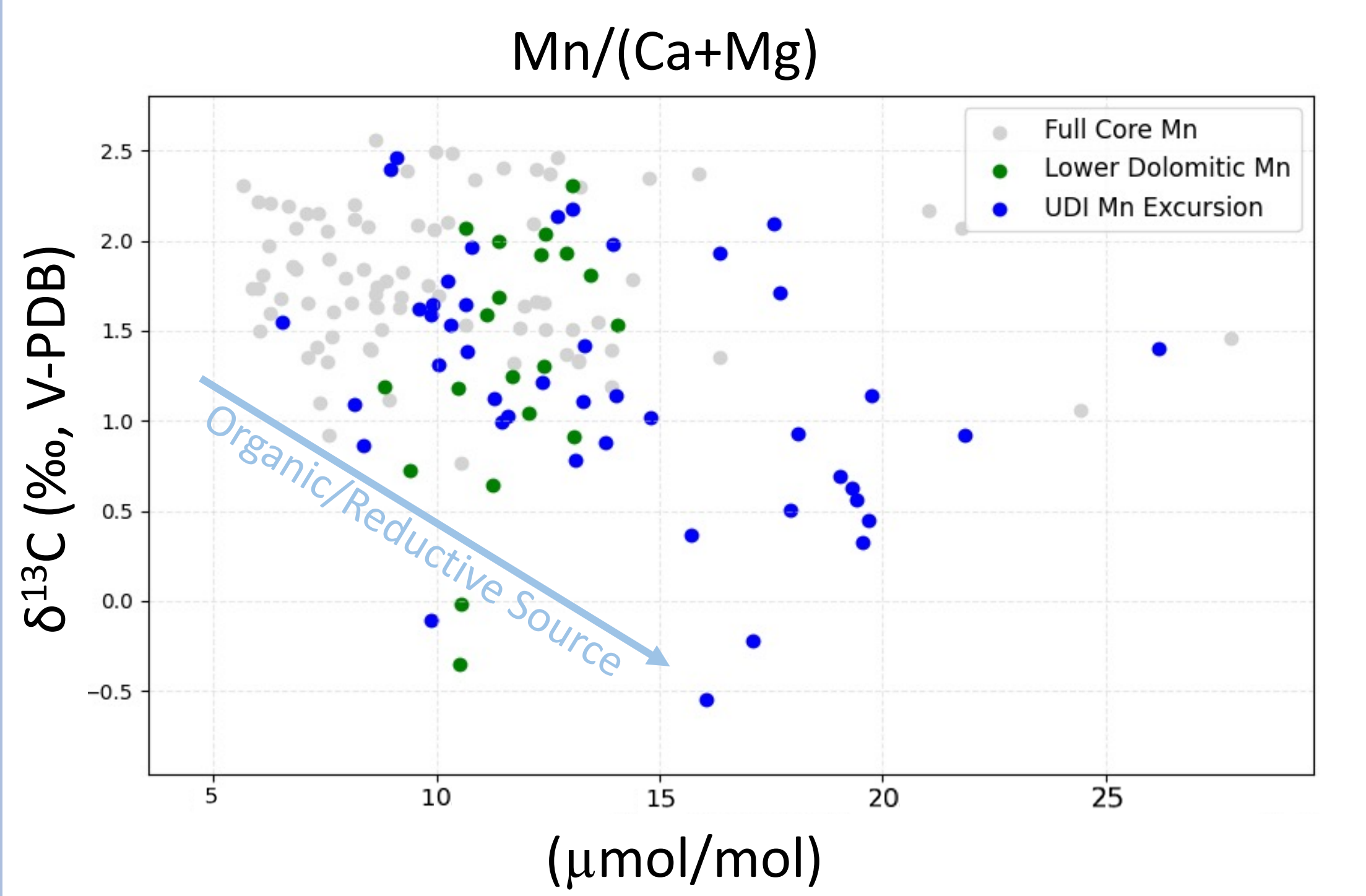
Lower Dolomitic Interval



Middle Calcareous Interval



Upper Dolomitic Interval



- Vuggy Cream Dolomite
- Orange sucrosic and creamy massive dolomite
- Chalky

Key Points

- Excursion may be a result of reducing conditions which led to higher Mn incorporation
- Negative $\delta^{13}\text{C}$ most likely from organic source
- Dolomitizing conditions of upper interval may be variable due to terrestrial influenced diagenetic fluids
- Mn responded differently to each dolomitizing processes occurring

Key Points

- Negative correlation between U/(Ca+Mg) and Cr/(Mg+Ca) points to change from suboxic to oxic environment. [1] [4]
- $\delta^{44/40}\text{Ca}$ suggests closed to open system change, values become closer to sea water, positive correlation with Uranium suggests this as well
- Concentrations of U and Cr most likely driven by diagenesis and system openness

Key Points

- Vanadium from shells can be used for reconstructing past oceanic and redox conditions [2] [3]
- Vanadium did not correlate with other redox trace elements, but it did with what would seem to be bioclastic grains in samples as shown by PDF graphs

Refs:

[1] Coyte, Rachel M., and Avner Vengosh. "Factors controlling the risks of co-occurrence of the redox-sensitive elements of arsenic, chromium, vanadium, and uranium in groundwater from the Eastern United States." *Environmental Science & Technology*, vol. 54, no. 7, 2020, pp. 4367–4375, <https://doi.org/10.1021/acs.est.9b06471>.
[2] Hastings, David W., Steven R. Emerson, and Alan C. Mix. "Vanadium in foraminiferal calcite as a tracer for changes in the areal extent of reducing sediments." *Paleoceanography*, vol. 11, no. 6, 1996, pp. 665–678, <https://doi.org/10.1029/96pa01985>.
[3] Hastings, David W., Steven R. Emerson, Jonathan Erez, et al. "Vanadium in foraminiferal calcite: Evaluation of a method to determine Paleo-seawater vanadium concentrations." *Geochimica et Cosmochimica Acta*, vol. 60, no. 19, 1996, pp. 3701–3715, [https://doi.org/10.1016/0016-7037\(96\)00196-2](https://doi.org/10.1016/0016-7037(96)00196-2).
[4] Higgins, J.A., et al. "Mineralogy, early marine diagenesis, and the chemistry of shallow-water carbonate sediments." *Geochimica et Cosmochimica Acta*, vol. 220, 2018, pp. 512–534, <https://doi.org/10.1016/j.gca.2017.09.046>.
[5] Maliva, R. G., et al. "Insights into the dolomitization process and porosity modification in Sucrosic Dolomites, Avon Park Formation (Middle Eocene), east-central Florida, U.S.A." *Journal of Sedimentary Research*, vol. 81, no. 3, 2011, pp. 218–232, <https://doi.org/10.2110/jsr.2011.18>.